EXPERIMENT NO – 2

Aim: Estimation of copper from PCB

Theory:

Printed circuit board is an essential part of all electrical and electronic equipment's. It is composed of polymers, ceramics and metals which all are come under Hazardous and non-hazardous categories. About 40% heavy metals present in it which are turned into hazardous residue if the discarded pcbs are not disposing properly. Metal compositions in PCBs are varies according to the manufacturer and the year of its manufacturing and technology Table 1 shows the composition of PCB.

Metals	Availability (wt%)
Copper	30.57
Aluminium	11.69
Zinc	1.86
Tin	7.3
Nickel	1.58
Iron	15.21
Lead	6.71

A solution of copper is made by dissolution of the PCB sample in Nitric acid. Oxides of Nitrogen are destroyed by boiling with Urea. Excess of acid is neutralized by adding Ammonia. The solution is changed to weak acidic medium by adding Acetic acid. Potassium Iodide is added. Iodine is liberated by the cupric ions. Then the solution is titrated against Sodiumthiosulphate solution using starch as an indicator. The amount of Sodium-thiosulphate consumed is a measure of the amount of copper present. The principle of this experiment is iodometric titration.

$$Cu + HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$

$$2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$$

$$HNO_2 + NH_2CONH_2 \rightarrow N_2 + CO_2 + H_2O$$

$$Cu(NO_3)_2 + NH_4OH \rightarrow NH_4NO_3 + Cu(OH)_2$$

$$Cu(OH)_2 + CH_3COOH \rightarrow Cu(CH_{3C}OO)_2 + H_2O$$

$$2Cu^{2+} + 4KI \rightarrow Cu_2I_2 + 4K^+ + I_2$$

$$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$$

Description:

Part-A: Preparation of standard Sodium thiosulphate solution

Weigh accurately the given $Na_2S_2O_3$ and transfer it into a 100 cm^3 standard flask using a funnel. Add small amount of distilled water to dissolve the salt and make it up to the mark by adding some more distilled water. Shake it well to get uniform concentration. Rinse the burette and fill the burette with the prepared standard sodium thiosulphate solution.

Normality of $Na_2S_2O_3$ = (weight of $Na_2S_2O_3$)/ gram equivalent weight of $Na_2S_2O_3$ (248.17)

Part-B: Preparation of copper solution and estimation of copper

Weigh exactly the given sample of PCB into a clean 250 cm^3 conical flask. Add Conc. Nitric acid drop by drop until the Brass dissolves completely and filter it into a 250 cm^3 volumetric flask. Make it up to the mark using distilled water.

Take 25 cm^3 of the prepared brass solution. Add one spatula urea and 1 tt of distilled water to the solution and heat it to boiling and cool it. Add Ammonium Hydroxide drop by drop until a pale blue precipitate is obtained. Dissolve the precipitate by adding dil. Acetic acid drop by drop, followed by ¼ tt of Acetic acid and 1 tt of 10% KI solution. Add a test tube of distilled water. Swirl the conical flask thoroughly for about 30 seconds. Titrate the liberated iodine against standard Sodiumthiosulphate solution taken in the burette until the solution becomes pale yellow or cream colored. Add about 2 cm^3 of freshly prepared starch solution as an indicator. Continue the titration by adding Sodiumthiosulphate solution **strictly drop by drop** until the dark blue coloration disappears, leaving behind a white ppt.
Repeat Part- B two more times. Calculate the percentage of copper present in the Brass samples by taking one titer value.

Part-C: Calculation

Weight of PCB sample dissolved in 250 cm^3 of solution = 'W' g (W = 5.0 g)

Normality of sodium thiosulphate = 'Y'

Volume of sodium thiosulphate = 'X' cm^3 (from titration)

1000 cm^3 1N sodium thiosulphate = 63.54g of copper

Therefore 'X' cm^3 of 'Y' N sodium thiosulphate = $\frac{63.54*X*Y}{1000}$ g of copper

= 'A' g of Cu

250cm^3 of prepared PCB solution contains = '10A' g of Cu

Model Procedure:

Part A:- Making of $Na_2S_2O_3$ salt solution as titrant by first measuring salt with weighing scale & fill in standard flask with distilled water and pour it in burette.

Part B :- 250 cm^3 of PCB soln & 1 spatula of urea. Then add NH_4OH drop by drop till pale blue ppt and ½ tt. of CH_3COOH, 1 tt of KI & 1 tt of distilled water. Titrate with $Na_2S_2O_3$ soln till pale yellow colour - Then add starch to get dark blue colour and continue titration

till solution becomes colourless.

Model Calculation:

Weight of PCB sample dissolved in 250 cm^3 of solution = 'W' g

Normality of sodium thiosulphate = 'Y'

Volume of sodium thiosulphate = 'X' cm^3

1000 cm^3 of 1 N sodium thiosulphate = 63.54 g of Cu.

Therefore 'x' cm^3 of 'y' N $Na_2S_2O_3$

$$= \frac{63.54 \times x \times y}{1000} \text{ g of Cu}$$

$$= A \text{ g}$$

250 cm^3 of prepared PCB soln contains = '10A' g of Cu

Observation and Calculation:

Part-A: Preparation of standard Sodium thiosulphate solution

Weight of the Weighing Bottle + $Na_2S_2O_3$	6.034	g
Weight of the Weighing Bottle	4.365	g
Weight of $Na_2S_2O_3$	1.669	g

$$\text{Normality of } Na_2S_2O_3 \text{ solution} = \frac{\text{Weight of } Na_2S_2O_3 \text{ dissolved in 100 } cm^3 \times 10}{\text{Gram equivalent weight of } Na_2S_2O_3 \text{ salt (248.17)}} = \frac{1.669 \times 10}{248.17}$$

$$= 0.067 \text{ N}$$

Part-B: Estimation of Copper in Brass solution

Burette readings (in cm^3)	Trial-I	Trial-II	Trail-III	Expected Value by teacher
Final	6.5	4.4	6.5	
Initial	0	0	0	6.7
Volume of $Na_2S_2O_3$ run down	6.5	4.4	6.5	

Part-C: Calculation

Weight of PCB sample dissolved in 250 cm^3 of sol^n = 5g

Normality of $Na_2S_2O_3$ = 0.067

Volume of $Na_2S_2O_3$ = 6.5 cm^3

1000 cm^3 of 1N $Na_2S_2O_3$ = 63.54 g of Cu

∴ 6.5 cm^3 of 0.067 N $Na_2S_2O_3$ = $\frac{63.54 \times 6.5 \times 0.067}{1000}$

= 0.027 g Cu

∴ 250 cm^3 of prepared PCB sol^n contains

= 0.27 g of Cu

Inference:

- The amount of Iodine liberated in the rx^n is the measure of quantity of Cu in PCB sol^n.
- Finding the vol. of $Na_2S_2O_3$ we can estimate the amount of Cu present in 250 cm^3 of PCB sol^n.
- Colour change from dark yellow to pale yellow to white
- Starch is an adsorption indicator.

Relevance to Society & Environment:

- Estimation of copper can be used to check % of copper in e-waste like wires & PCB. and can help in recycling of copper.
- Helps in estimating copper in glasses

Report: Amount of Cu present in the given 5 g of PCB sample = 0.27 g of Cu

Evaluation of experiment - 2		
Components	Marks	
	Max	Obtained
Model Procedure & Calculation	7 + 5	12
Burette Reading & Execution	20 + 4	24
Inference & Societal Relevance	2 + 2	3
Total	40	39
Signature of Teacher		1/12/23

EXPERIMENT NO – 3

Aim: Determination of total acid content of the soft drinks using pH sensor

Theory:

The dissociation constant of a weak acid is denoted as K_a and it is given by Ostwald's dilution law.

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

$$K_a = \frac{[H^+]\,[CH_3COO-]}{[CH_3COOH]}$$

The strength of the acid is indicated by its dissociation constant. Dissociation constant is also a measure of the tendency of an acid to donate proton. Generally, K_a value of weak acids are very small. In order to avoid negative power appearing as K_a value, a new term called pK_a value is coined. If K_a is the dissociation constant of a weak acid, then

$$pK_a = -\log_{10} K_a$$

The pH of a weak acid solution and pK_a are related as per Henderson - Hasselbalch equation:

$$pH = pK_a + \log_{10}\frac{[salt]}{[acid]} \quad \ldots\ldots\ldots\ldots (1)$$

Addition of a strong base (Sodium Hydroxide) to a weak acid (Acetic acid) produces salt (Sodium Acetate). At equivalence point (end point) all the acid is neutralized. At half equivalence point, half of the acid is neutralized, i.e., [salt formed] = [acid unreacted]. Therefore, equation (1) reduces to $pH = pK_a$. This indicates that at half equivalence point the pK_a of the weak acid is equal to pH. Therefore, **the pKa of a weak acid can also be defined as the pH at half equivalence point.**

During an acid-base titration, initially the pH value of the solution increases gradually and increases more rapidly thereafter till the equivalence point is reached. Once the equivalence point is crossed, pH increases by small amounts. The variation of pH is measured using combined glass electrode.

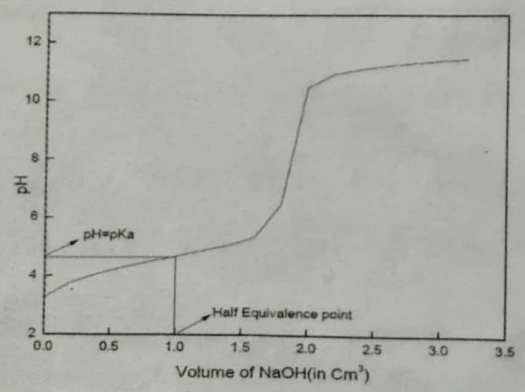
Description:

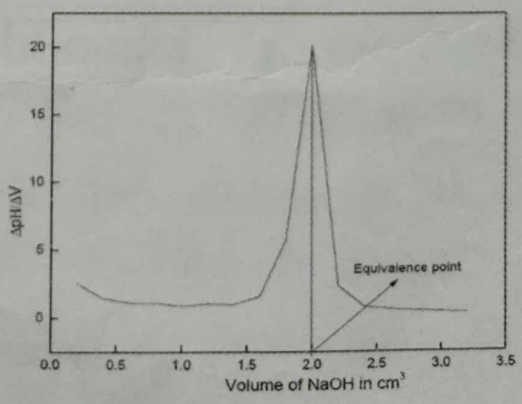
Pipette out 50 cm^3 of the given soft drinks (weak acid) into a clean 250 cm^3 beaker. Insert glass electrode - calomel electrode assembly into it and connect it to pH meter. Measure the pH of the acid. Fill the burette with the given sodium hydroxide. In the beginning add 0.2 cm^3 of NaOH at a time and stir the mixture. Measure the pH. Continue adding 0.2 cm^3 of NaOH & measure the pH regularly after each addition until the increase in pH is comparatively large. Take a few more reading by adding 0.2 cm^3 of NaOH.

Determine the pKa of the given weak acid from the two graphs:

(i) $\frac{\Delta pH}{\Delta V}$ against volume of NaOH added.

(ii) pH against volume of NaOH added





Calculation:

The experiment is divided into two parts.

1. Determination of equivalence point
2. Determination of pKa. (Using table-1 corresponding to pKa find out which weak acid is present)

Table-1

Acid	pKa	Gram equivalent weight
CH_3COOH (acetic)	4.7	60/1 = 60
HCOOH (formic)	3.7	46/1 = 46
H_2CO_3 (carbonic)	6.3	62/2 = 31
$H_2C_2O_4$ (oxalic)	1.2	126/2 = 63

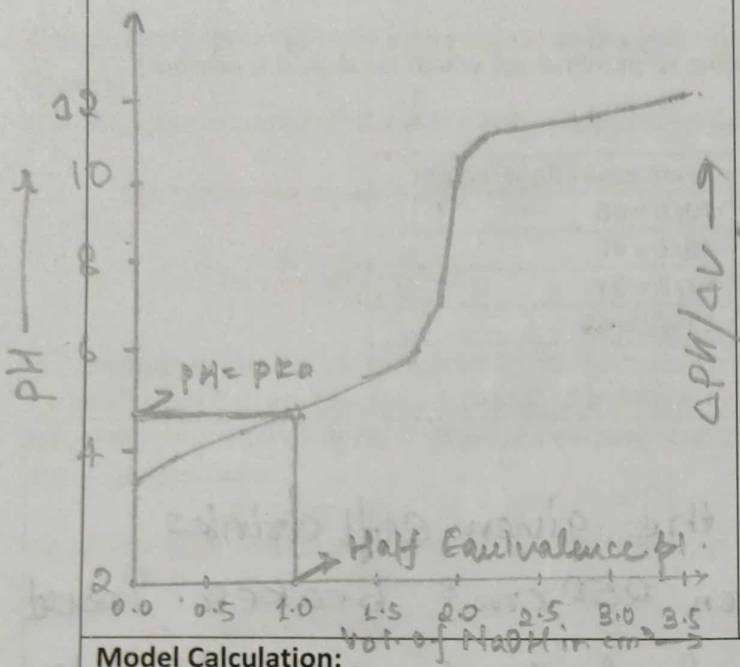
Model Procedure /Flow Chart:

Pipette out 50 cm^3 of the given soft drinks (weak acid) into a clean 250 cm^3 beaker. Insert glass - electrode - calomel electrode assembly into it and connect it to pH meter. Measure the pH of the acid. Fill the burette with the sodium hydroxide. In the beginning add 0.2 cm^3 of NaOH at a time & stir the mixture. Measure the pH. Continue adding 0.2 cm^3 of NaOH & measure the pH regularly until the increase in pH is comparatively large. Take a few more readings by adding 0.2 cm^3 of NaOH. Determine the pKa of the given weak acid from two graphs:

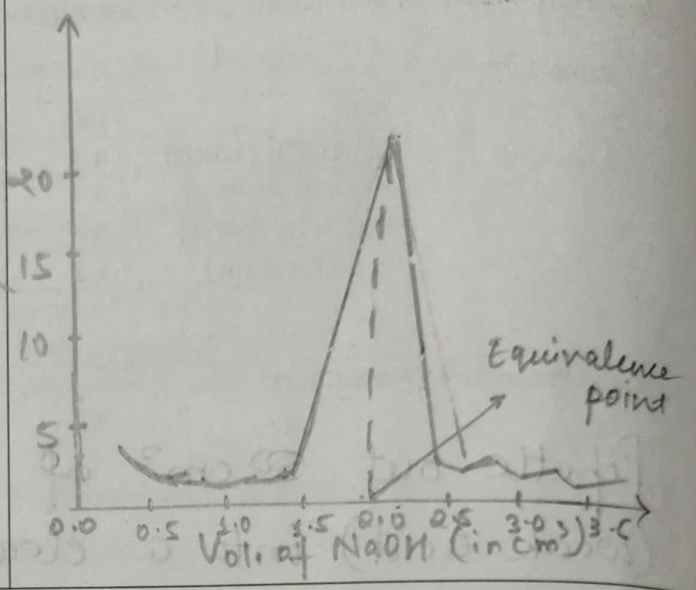
(i) $\frac{\Delta pH}{\Delta V}$ against volume of NaOH added

(ii) pH against volume of NaOH added.

Model graph 1:



Model graph 2:



Model Calculation:

1) Determine end point or equivalence point.

2) Determine pka

From equivalence point, we find N of weak acid.

$$(NV)_{acid} = (NV)_{NaOH}$$

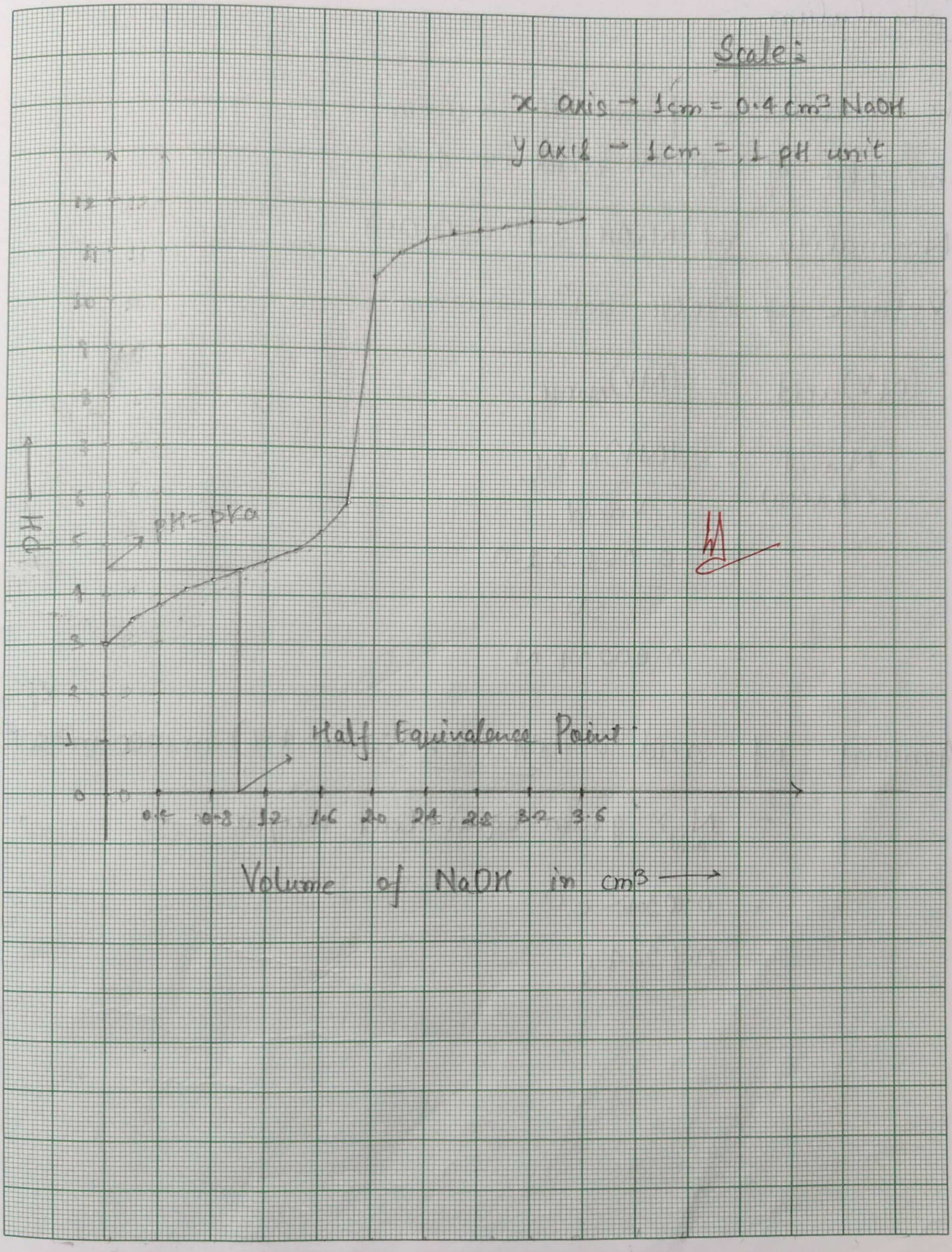
$$N_{acid} = \frac{(NV)_{NaOH}}{V_{acid}}$$

Amount of acid in 1000 cm^3 of sol^n

= N acid × gm eq. weight of acid.

Tabulation:

Volume of NaOH added. in cm^3	pH	ΔpH	ΔV	$\frac{\Delta pH}{\Delta V}$
0.0	3 ~ 3.01			
0.2	3.5 ~ 3.46	0.45	0.2	2.25 ~ 2.3
0.4	3.8 ~ 3.83	0.37	0.2	1.85 ~ 1.9
0.6	4.1 ~ 4.06	0.23	0.2	1.15 ~ 1.2
0.8	4.3 ~ 4.27	0.21	0.2	1.05 ~ 1.1
1.0	4.5 ~ 4.51	0.24	0.2	1.20 ~ 1.2
1.2	4.7 ~ 4.71	0.20	0.2	1.00 ~ 1
1.4	4.9 ~ 4.92	0.21	0.2	1.05 ~ 1.1
1.6	5.3 ~ 5.26	0.34	0.2	1.70 ~ 1.7
1.8	5.9 ~ 5.88	0.62	0.2	3.10 ~ 3.1
2.0	10.5 ~ 10.52	4.64	0.2	23.2 ~ 23.2
2.2	11 ~ 11.03	0.51	0.2	2.55 ~ 2.6
2.4	11.3 ~ 11.26	0.23	0.2	1.15 ~ 1.6
2.6	11.4 ~ 11.41	0.15	0.2	0.75 ~ 0.8
2.8	11.5 ~ 11.51	0.10	0.2	0.50 ~ 0.5
3.0	11.6 ~ 11.61	0.10	0.2	0.50 ~ 0.5
3.2	11.7 ~ 11.68	0.07	0.2	0.35 ~ 0.4
3.4	11.7 ~ 11.74	0.06	0.2	0.30 ~ 0.3
3.6	11.8 ~ 11.80	0.06	0.2	0.30 ~ 0.3


Scale:
x axis → 1cm = 0.4 cm³ NaOH
y axis → 1cm = 1 pH unit
pH
pH = pKa
Half Equivalence Point
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
Volume of NaOH in cm³ →

Calculation :-

The equivalence point is $2.0\ cm^3$

The pka of given acid is 4.5.

Normality of NaOH sol^n = 0.051

Volume of NaOH sol^n = $2.0\ cm^3$

$$(NV)_{acid} = (NV)_{NaOH}$$

$$N_{acid\ (CH_3COOH)} = \frac{(NV)_{NaOH}}{V_{acid}}$$

$$= \frac{0.051 \times 2.0}{50}$$

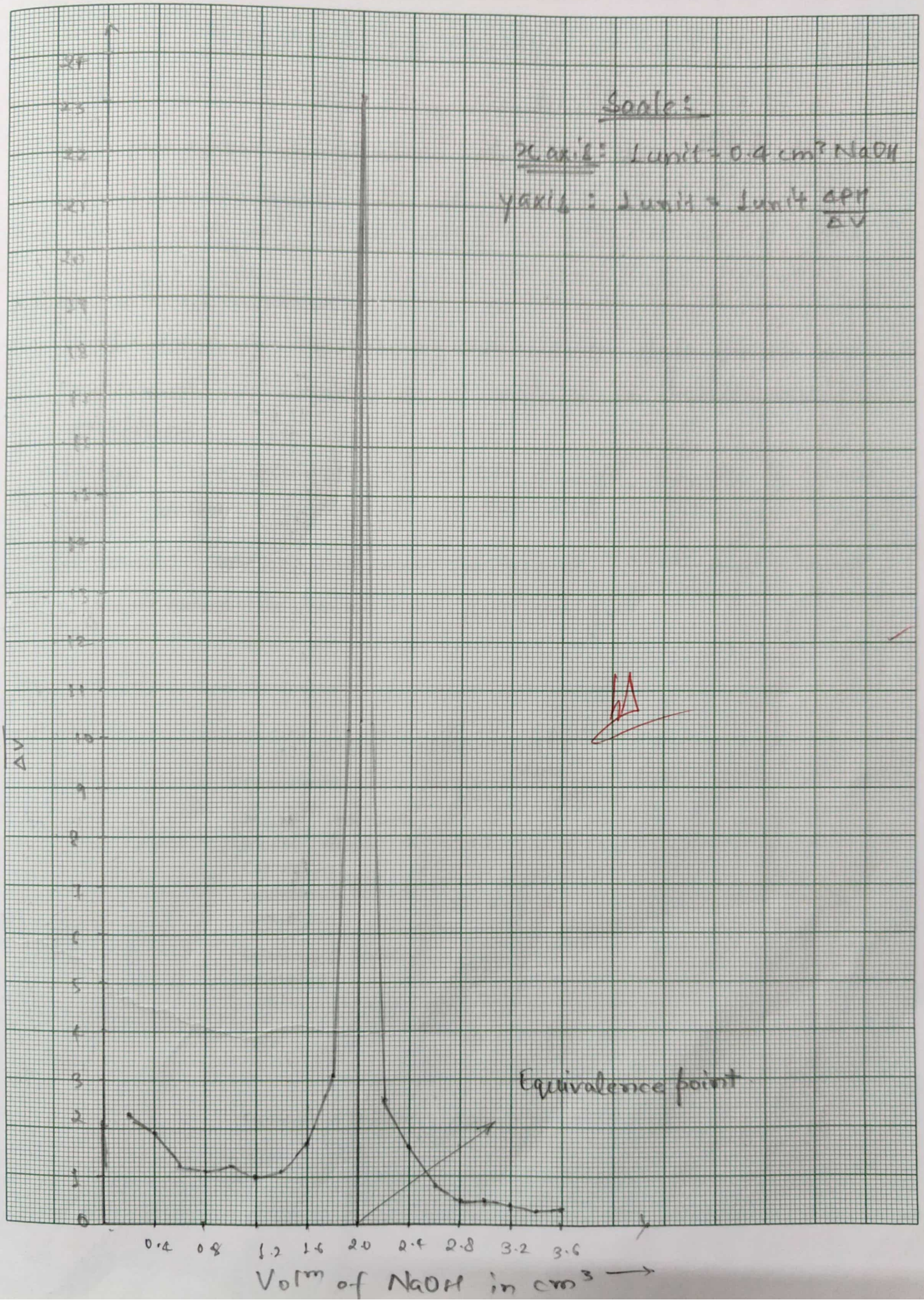
$$= 0.00204\ N$$

Amount of acid i.e. CH_3COOH ($\because$ pka $\simeq$ 4.7) in $1000\ cm^3$ sol^n

$$= N_{acid} \times \text{gram equivalent weight}$$

$$= 0.00204 \times 60$$

$$= 0.1224\ g$$


Scale:
x axis: 1 unit = 0.4 cm³ NaOH
y axis: 1 unit = 1 unit
ΔpH / ΔV
Equivalence point
0.4
0.8
1.2
1.6
2.0
2.4
2.8
3.2
3.6
Volᵐ of NaOH in cm³

Inference:

- In this experiment, we measure the pH of soft drink.
- The pH between 2.5-3.5 is highly acidic.
- It provides a general idea of the pH levels of the given soft drink.
- Wee see a huge spike in pH because OH^- ions increase after all acidic H^+ ions have reacted.

Relevance to Society & Environment:

- Understanding the acid content in soft drink is relevant to society as it relates to health considerations.
- It can have adverse effect on dental health & overall well being if the pH is too low.
- Accurate measurements contribute to informed choices.

Report: The acid content of the given soft drink = 0.1224 g

Evaluation of experiment - 3		
Components	**Marks**	
	Max	**Obtained**
Model Procedure, Model Graph & Calculation	9 + 7	16
Equivalence Point & Execution	16 + 4	20
Inference & Societal Relevance	2 + 2	3
Total	40	39
Signature of Teacher		1/12/23

EXPERIMENT NO – 4

Aim: Potentiometric estimation of iron

Theory:

Nernst equation gives the relation between the potential at an electrode and the concentration of the active species in the solution.

For the reaction,

oxidized form + ne⁻ ⟷ reduced form,

Nernst equation can be written as

$$E = E^{\circ} + \frac{0.0591}{n} \log \frac{[oxidized\ form]}{[reduced\ form]}$$

Where E° is the standard electrode potential and the terms within the brackets represent their concentrations.

It is evident from the equation that the potential depends upon the concentrations of the oxidised and the reduced species in the solution. When a reducing agent in solution is oxidized using an oxidizing agent, the concentrations and hence the potential changes. Potentiometry essentially involves the measurement of change in potential as and when a species in solution is oxidized or reduced. When a titration of a reducing agent is carried out against an oxidising agent, the potential gradually changes in the beginning but changes rapidly near the end point. A plot of change in potential against volume reveals a sudden change in potential at the equivalence point. This sudden change in the present experiment can be explained as follows. The reactions that take place when dichromate is added to ferrous solution are:

$$Fe^{2+} \rightarrow Fe^{3+} + e^{-}$$

$$Cr_2O_7^{2-} + 14H^{+} + 6e^{-} \rightarrow 2Cr^{3+} + 7H_2O$$

The potential of the system before the equivalence point is given by

$$E = E^{O} + \frac{0.0591}{1} \log \frac{[Fe^{3+}]}{[Fe^{2+}]}$$

i.e., the potential is dependent on the concentration of Fe^{2+} and Fe^{3+} ions. The potential of the solution will be around 0.75V. At the equivalence point the potential is decided by the concentrations of Fe^{2+}, Fe^{3+}, $Cr_2O_7^{2-}$ and Cr^{3+} ions. The potential at equivalence point is given by

$$E = \frac{E^{0}_{Fe^{3+}/Fe^{2+}} + E^{0}_{Cr^{6+}/Cr^{3+}}}{2}$$

$$= \frac{0.75V + 1.33V}{2} = 1.04V$$

Beyond the equivalence point as no ferrous ions exist, the potentials is determined by the concentration of $Cr_2O_7^{2-}$ and Cr^{3+} ions and it is given by

$$E = E^{O} + \frac{0.0591}{6} \log [Cr^{6+}]/ [Cr^{3+}]$$

$$= 1.33V + \frac{0.0591}{6} \log [Cr^{6+}]/ [Cr^{3+}]$$

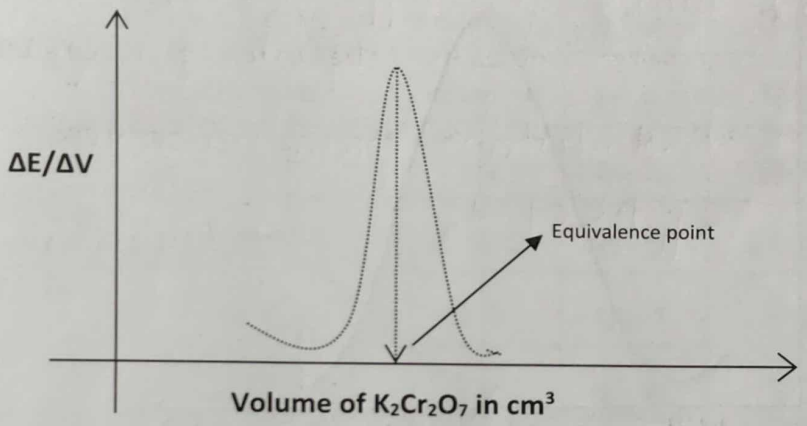
therefore, an abrupt increase in the potential is observed at the equivalence point. This increase marks the end point of the titration. Potential change at an electrode cannot be directly measured. The electrode at which the potential changes with concentration (indicator electrode) is connected to another electrode whose potential remains constant [ex:-calomel electrode (Standard reference electrode)], to form a cell.

$$EMF = E_{cathode} - E_{anode}.$$

As the potential of the indicator electrode changes, the EMF of the cell also changes. It is the change in EMF that is measured during a potentiometric titration.

Description:

FAS solution is prepared by dissolving known amount of FAS in 1 tt of dilute Sulphuric acid and little amount of water. Using the dispenser take 25 cm^3 of rust solution into a clean beaker. Add 1 test tube of dil. Sulphuric acid. Dip the electrode assembly into the solution and connect to a potentiometer. Measure the potential. Add 0.2 cm^3 of Potassium dichromate from a burette. Stir the solution well and measure the potential. Continue the process till the potential shows a tendency to increase rapidly. Now add dichromate in increments of 0.2 cm^3 and measure the potential after each addition. Plot a graph of $\Delta E/\Delta V$ against volume of dichromate added as shown in the figure and find out the end point. Calculate the normality of the ferrous solution and determine the amount of iron in the given volume.



Calculation:

Volume of $K_2Cr_2O_7$ required for the reaction = Vcm^3 (From graph),

$$N_{iron} = \frac{NV_{K2Cr2O7}}{V_{iron}}$$

Amount of iron presented in 1000 cm^3 of its solution = N_{iron} x gram equivalent weight of Iron = ...B g

Model Procedure /Flow Chart:

FAS solution (25 ml) is taken in a beaker & 1 test tube of H_2SO_4 is added. We take standard $K_2Cr_2O_7$ in a microburette.

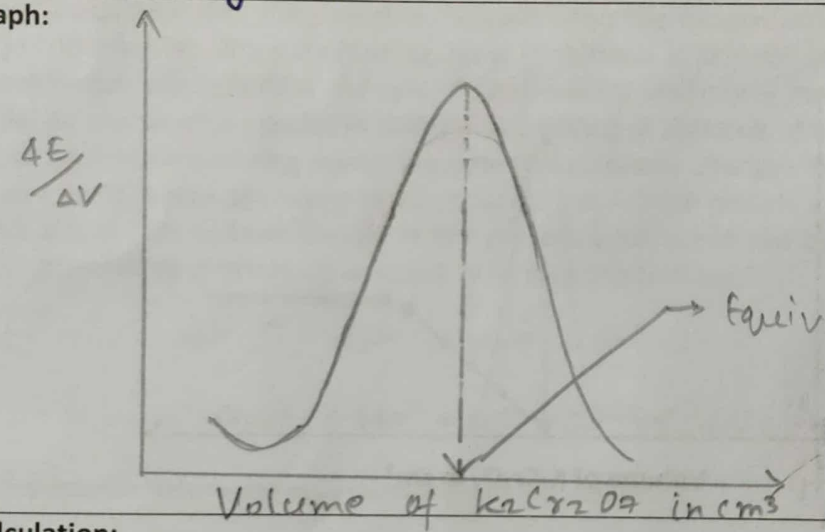
& is used for titration. FAS (known volume) is titrated against std. sol^n of $K_2Cr_2O_7$ (known volume & known conc^n) to find the concr of iron in FAS by redox rx^n titration.

Setup: $Fe^{2+} \rightleftharpoons Fe^{3+} + e^-$ $\quad (N_1V_1)_{FAS} = (N_2V_2)_{K_2Cr_2O_7}$

$Cr^{6+} + 3e^- \rightleftharpoons Cr^{3+}$

Electrode is dipped in FAS sol^n & connected to potentiometer. We take the reading of potential after every 0.2 cm^3 of $K_2Cr_2O_7$. Then Normality is found after a peak rise in voltage & taken 2-3 readings after it.

Model graph:



Model Calculation:

Volume of $K_2Cr_2O_7$ required for the rx^n = V cm^3 (from graph)

N = Normality of $K_2Cr_2O_7$

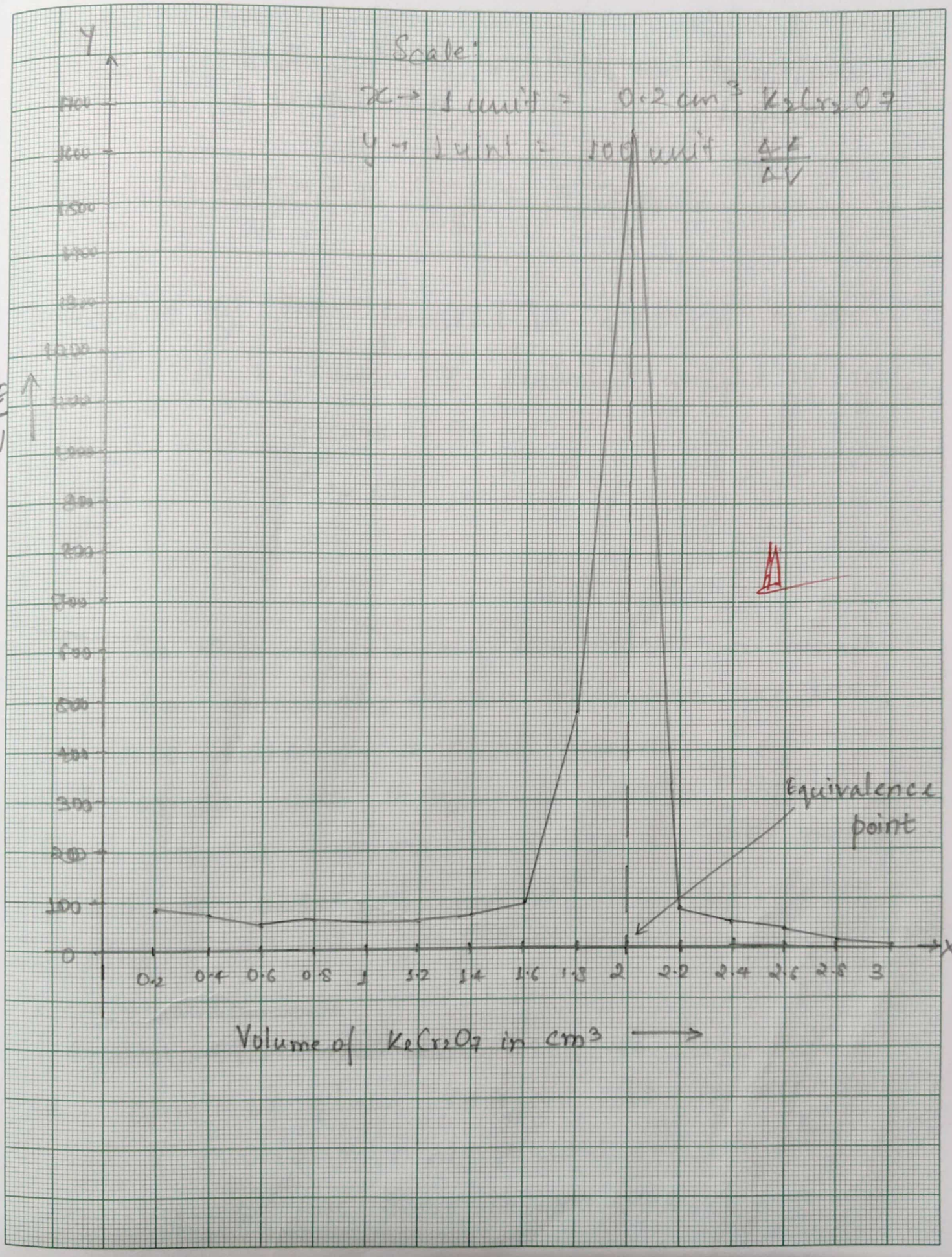
$$N_{iron} = \frac{NV_{K_2Cr_2O_7}}{V_{iron}}$$

Amount of iron present in 1000 cm^3 of its sol^n

= N_{iron} × gm equivalent weight of iron.

Tabulation:

Volume of $K_2Cr_2O_7$ added in cm^3	Potential (E) In mV	ΔE	ΔV	ΔE/ΔV
0.0	264			
0.2	281	17	0.2	85
0.4	295	14	0.2	70
0.6	305	10	0.2	50
0.8	317	12	0.2	60
1.0	328	11	0.2	55
1.2	339	11	0.2	55
1.4	353	14	0.2	70
1.6	372	19	0.2	95
1.8	467	95	0.2	475
2.0	799	332	0.2	1660
2.2	815	16	0.2	80
2.4	826	11	0.2	55
2.6	834	8	0.2	40
2.8	837	3	0.2	15
3.0	842	5	0.2	25
3.2				
3.4				
3.6				


Y
Scale:
X → 1 unit = 0.2 cm³ K₂Cr₂O₇
Y → 1 unit = 100 unit ΔE/ΔV
ΔE/ΔV
Equivalence point
0
100
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
2.4
2.6
2.8
3
X
Volume of K₂Cr₂O₇ in cm³ ⟶

Calculation:

Volume of $K_2Cr_2O_7$ required for the reaction = 2 cm^3

$$N_{iron} = \frac{0.05 \times 2}{25} = \frac{0.1}{25} = 0.004N$$

Amount of iron present in 1000 cm^3 of its soln is = N_{iron} × gm equivalent weight of iron

$= 0.004 \times 55.85$

$= 0.2234$ gm

Inference: In this experiment the titration result are so accurate as no indicators are used. In this titration a small quantity of substances are required. During titration, there is a sudden increase in the potential which is observed to be the equivalence point.

Relevance to Society & Environment:

Potentiometric method is used to determine the conc^n of given reagent. It is economical & environmental process. It is also advantageous as there is no use of chemical

indicators rather electric potential across substances is measured making it more accurate.

Report: Amount of Iron present in 1000 cm^3 of FAS solution = 0.22 g

Evaluation of experiment - 4		
Components	**Marks**	
	Max	**Obtained**
Model Procedure, Model Graph & Calculation	9 + 7	16
Equivalence Point & Execution	16 + 4	~~19~~ 20
Inference & Societal Relevance	2 + 2	3
Total	40	39
Signature of Teacher		10/11/23

EXPERIMENT NO – 5

Aim: Conductometric estimation of strong acid with strong base

Theory:

Conductance of any solution (C) is the ease with which current flows through it. It is the reciprocal of resistance (R) offered by the solution.

$$C = 1/R$$

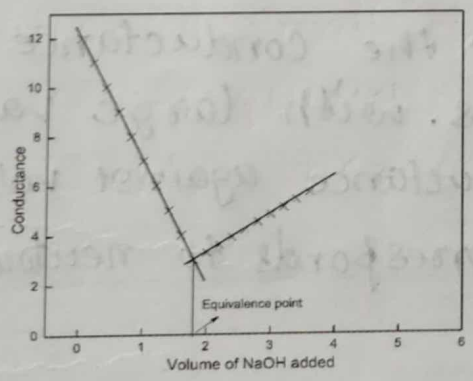
Conductance of a solution depends on number of ions, charge on the ions and also on the mobility of the ions. During titration, the conductance of the solution may change due to any of the above reasons. Such titrations in which conductance measurements are made use of in determining the equivalence point are called Conductometric titrations.

Among all Conductometric titrations, reaction involving strong acid (HCl) and strong base (NaOH) is the most common activity.

Hydrochloric acid is a strong electrolyte and would be completely in dissociated state. The conductance of this solution is the maximum, before the addition of NaOH. Upon addition of NaOH, highly mobile H^+ions of the acid are replaced by less mobile Na^+ ions.

$$H^+ + Cl^- + Na^+ + OH \longrightarrow {}^{-}Na^+ + Cl^- + H_2O$$

Hence the conductance would decrease steeply. The same trend would continue till all the H^+ ions available in the acid are completely reacted with the added NaOH. After the neutralization of all hydrogen ions of strong acid, sodium hydroxide solution will neutralize hydrogen ions of acetic acid. Now, the addition of NaOH would add more and more of Na^+ and OH^- ions. Due to relatively higher mobility of OH^- ions, the conductance of the solution increases again. A typical graph of conductance vs. volume of NaOH (added) is given below.



The intersection of two lines corresponds to the equivalence point corresponding to strong acid. From the volume of NaOH required to neutralize HCl, the amount of HCl is calculated.

Description:

Pipette out 50 cm^3 of acid into a clean 200 cm^3 beaker. Dip the conductivity cell in the acid, such that the two platinum electrodes are immersed completely. The cell is connected to the conductivity bridge. Measure the conductance. Take the given NaOH solution in a burette rinsed with the same solution. Rundown 0.2 cm^3 of NaOH solution into the beaker and shake well. Note down the conductance. Note down the conductance by

adding 0.2 cm^3 of NaOH each time to the beaker. Conductance decreases in the beginning and finally starts increasing by large value. Take around 8-10 readings on the increasing side. Draw a graph of conductance against volume of NaOH added. The equivalence point in the graph corresponds to the neutralization of HCl. Calculate the normality and amount of HCl present in 1000 cm^3 of its solution.

Calculation:

$$(NV)_{acid} = (NV)_{NaOH}$$

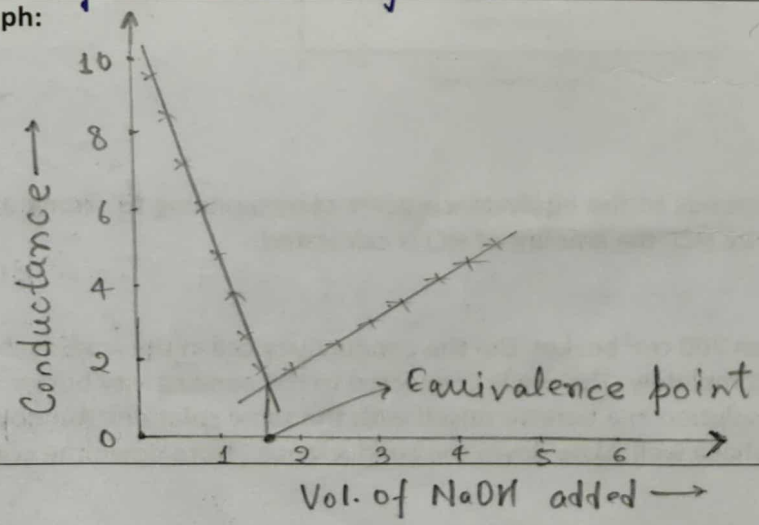
$$N_{acid} = \frac{(NV)_{NaOH}}{V_{acid}}$$

Amount of HCl or CH_3COOH present in 1000 cm^3 of its solution = N_{acid} x gram equivalent weight of acid

Model Procedure /Flow Chart:

Pipette out 50 cm^3 of acid into a 200 cm^3 beaker. Dip the conductivity cell in the acid such that two platinum electrodes are immersed completely. The cell is connected to connectivity bridge. Measure the conductance. Take the NaOH sol^n in burette rinsed with the same. Rundown 0.2 cm^3 of NaOH sol^n each time to the beaker. Take down the readings until you pipette out 4 cm^3 of NaOH sol^n. At first the conductance decreases and then increases with large values. Draw the graph of conductance against vol. of NaOH. The equivalence point corresponds to neutralisation of HCl.

Model graph:



Model Calculation:

$$(NV)_{HCl} = (NV)_{NaOH}$$

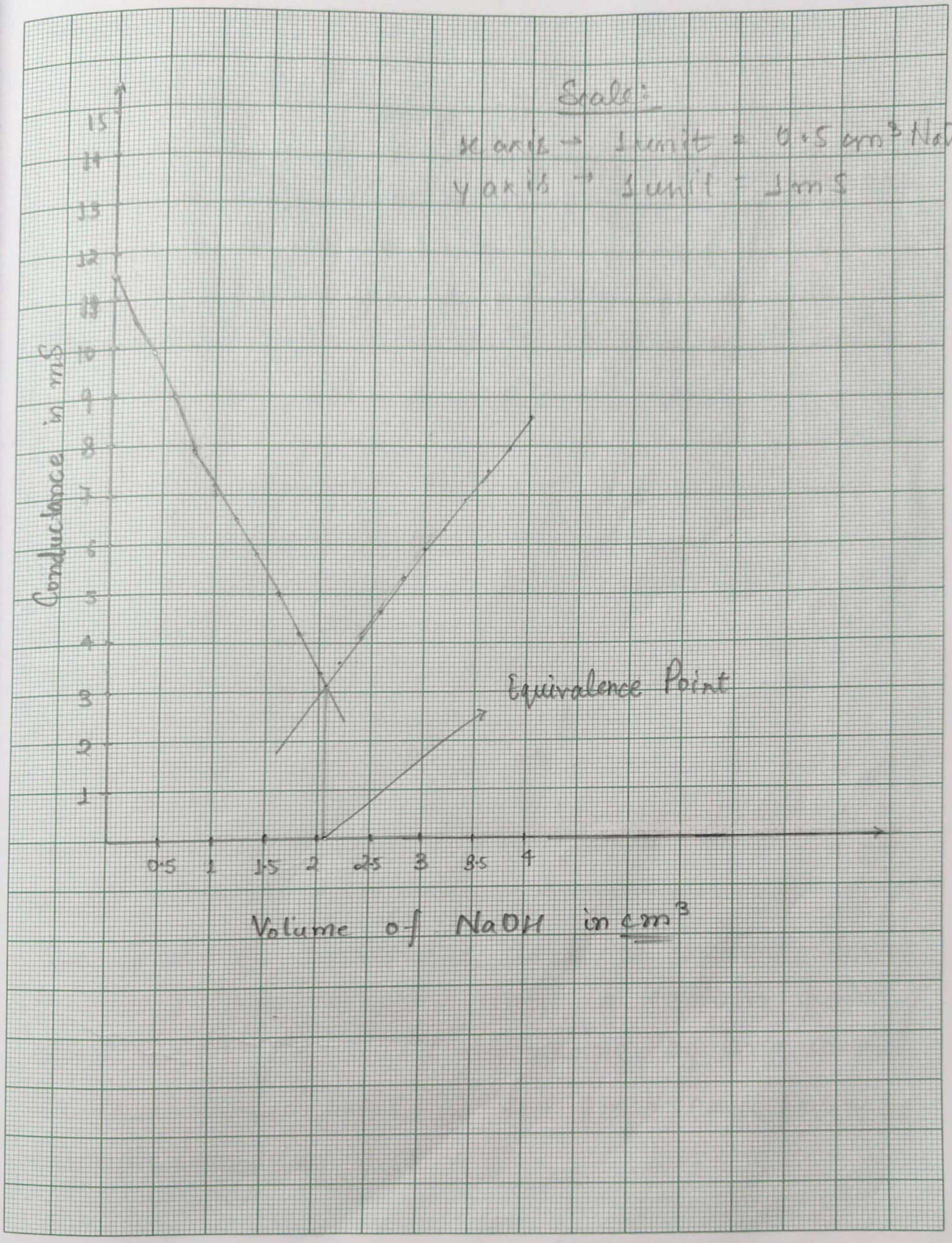
$$N_{HCl} = \frac{(NV)_{NaOH}}{V_{HCl}}$$

Amount of HCl present in 1000 cm^3 of its soln = N_{HCl} x gram equivalent weight of acid.

Tabulation:

Volume of NaOH in cm^3	Conductance in mS
0.0	11.4
0.2	10.6
0.4	9.9
0.6	9.0
0.8	7.9
1.0	7.2
1.2	6.5
1.4	5.8
1.6	5.0
1.8	4.2
2.0	3.4
2.2	3.6
2.4	4.1
2.6	4.8
2.8	5.4
3.0	5.9

Volume of NaOH in cm^3	Conductance in mS
3.2	6.3
3.4	6.9
3.6	7.5
3.8	7.9
4.0	8.6
4.2	
4.4	
4.6	
4.8	
5.0	
5.2	
5.4	
5.6	
5.8	
6.0	


Scale:
x axis → 1 unit = 0.5 cm³ NaOH
y axis → 1 unit = 1 mS
Conductance in mS
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0.5
1
1.5
2
2.5
3
3.5
4
Equivalence Point
Volume of NaOH in cm³

Calculation:

$$(NV)_{HCl} = (NV)_{NaOH}$$

$$N_{HCl} = \frac{(NV)_{NaOH}}{V_{HCl}} = \frac{0.049 \times 2.1}{50} = 0.0020N$$

Amount of HCl present in 1000 cm^3 of its sol^n =

= N_{HCl} × gm equivalent of HCl

= 0.0020 × 36.5

~~= 0.07665 gm~~

= 0.073 gm

Inference:

The value of conductance tends to decrease during the acid-base titration until a certain point, after which it starts increasing by large amount. The equivalence point in the graph corresponds to neutralisation of HCl.

Relevance to Society & Environment:

→ Conductometry can be used to examine the salinity of sea water & alkanity of fresh water.
→ This type of titration is used in food industries and food microbiologist to examine microorganism.
→ Conductometry is used in medical diagnosis to analyse biological fluids.

Report:

1. Normality of HCl = ~~0.0020~~ 0.073 N
2. Amount of HCl present in 1000 cm^3 of its solution = ~~0.0766~~ 0.073 g

Evaluation of experiment - 5		
Components	Marks	
	Max	Obtained
Model Procedure, Model Graph & Calculation	9 + 7	15
Equivalence Point & Execution	16 + 4	20
Inference & Societal Relevance	2 + 2	3
Total	40	38
Signature of Teacher		17/11/23

EXPERIMENT NO – 6

Aim: Conductometric estimation of acid mixture

Theory:

Conductance of any solution (C) is the ease with which current flows through it. It is the reciprocal of resistance (R) offered by the solution.

$$C = 1/R$$

Conductance of a solution depends on number of ions, charge on the ions and also on the mobility of the ions. During the course of a titration, the conductance of the solution may change due to any of the above reasons. Such titrations in which conductance measurements are made use of in determining the equivalence point are called Conductometric titrations.

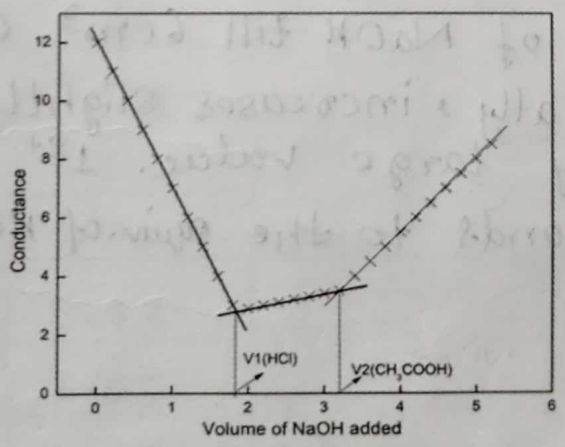
Among all Conductometric titrations, reaction involving strong acid (HCl)/ weak acid (CH_3COOH) and strong base (NaOH) is the most common activity.

Hydrochloric acid is a strong electrolyte and would be completely in dissociated state. The conductance of this solution is the maximum, before the addition of NaOH. Upon addition of NaOH, highly mobile H^+ions of the acid are replaced by less mobile Na^+ ions.

$$H^+ + Cl^- + Na^+ + OH \longrightarrow {}^-Na^+ + Cl^- + H_2O$$

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

Hence the conductance would decrease steeply. The same trend would continue till all the H^+ ions available in the acid are completely reacted with the added NaOH. After the neutralization of all hydrogen ions of strong acid, sodium hydroxide solution will neutralize hydrogen ions of acetic acid. Now, the addition of NaOH would add more and more of Na^+ and OH^- ions. Due to relatively higher mobility of OH^- ions, the conductance of the solution increases again. A typical graph of conductance vs. volume of NaOH (added) is given below.



The break in the curve corresponds to the two equivalence points corresponding to two acids. From the volume of NaOH required to neutralize HCl or CH_3COOH, the amount of HCl or CH_3COOH is calculated.

Description:

Pipette out 50$cm^3$of acid into a clean 200 cm^3beaker. Dip the conductivity cell in the acid, such that the two platinum electrodes are immersed completely. The cell is connected to the conductivity bridge. Measure the conductance. Take the given NaOH solution in a burette rinsed with the same solution. Rundown 0.2 cm^3 of

NaOH solution into the beaker and shake well. Note down the conductance. Note down the conductance by adding 0.2 cm^3 of NaOH each time to the beaker. Conductance decreases in the beginning, increases slightly and finally starts increasing by large value. Take around 8-10 readings on the increasing side. Draw a graph of conductance against volume of NaOH added. 1st equivalence point in the curve corresponds to the neutralization of HCl and the difference of two equivalence point corresponds to the neutralization of CH_3COOH acid. Calculate the normality and amount of HCl and CH_3COOH present in 1000 cm^3 of its solution.

Calculation:

$$(NV)_{acid} = (NV)_{NaOH}$$

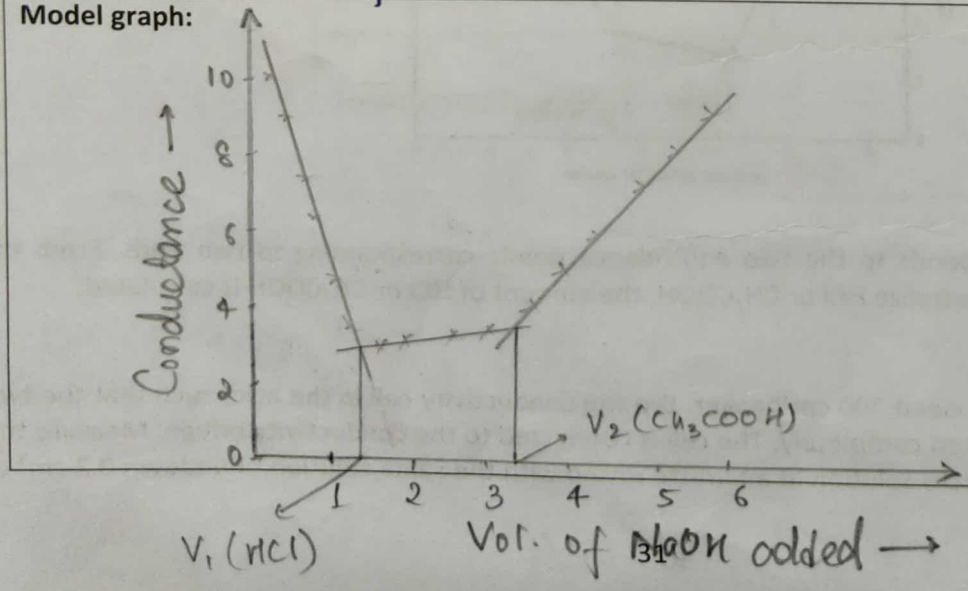
$$N_{acid} = \frac{(NV)_{NaOH}}{V_{acid}}$$

Amount of HCl or CH_3COOH present in 1000 cm^3 of its solution = N_{acid} x gram equivalent weight of acid

Model Procedure /Flow Chart:

Take a 200 cm^3 clean beaker & fill with 50 cm^3 of acid. Insert the conductivity cell in the beaker such that two platinum electrodes are immersed completely. The cell is connected to the conductivity bridge. Read the conductance. Take NaOH sol^n in the burette rinsed with same solution. Add 0.2 cm^3 of NaOH sol^n in beaker & shake well. Note down the conductance. Read down the conductance after every 0.2 cm^3 of NaOH till 6 cm^3. Conductance decreases initially, increases slightly then starts increasing by large value. 1st equivalence point corresponds to the equiv. of HCl & diff of two gives

Model graph:



Model Calculation:

$$(NV)_{acid} = (NV)_{NaOH}$$

$$N_{acid} = \frac{(NV)_{NaOH}}{V_{acid}}$$

Amount of HCl or CH_3COOH present in 1000 cm^3 of its solution = N_{acid} × gram equivalent weight of acid.

Tabulation:

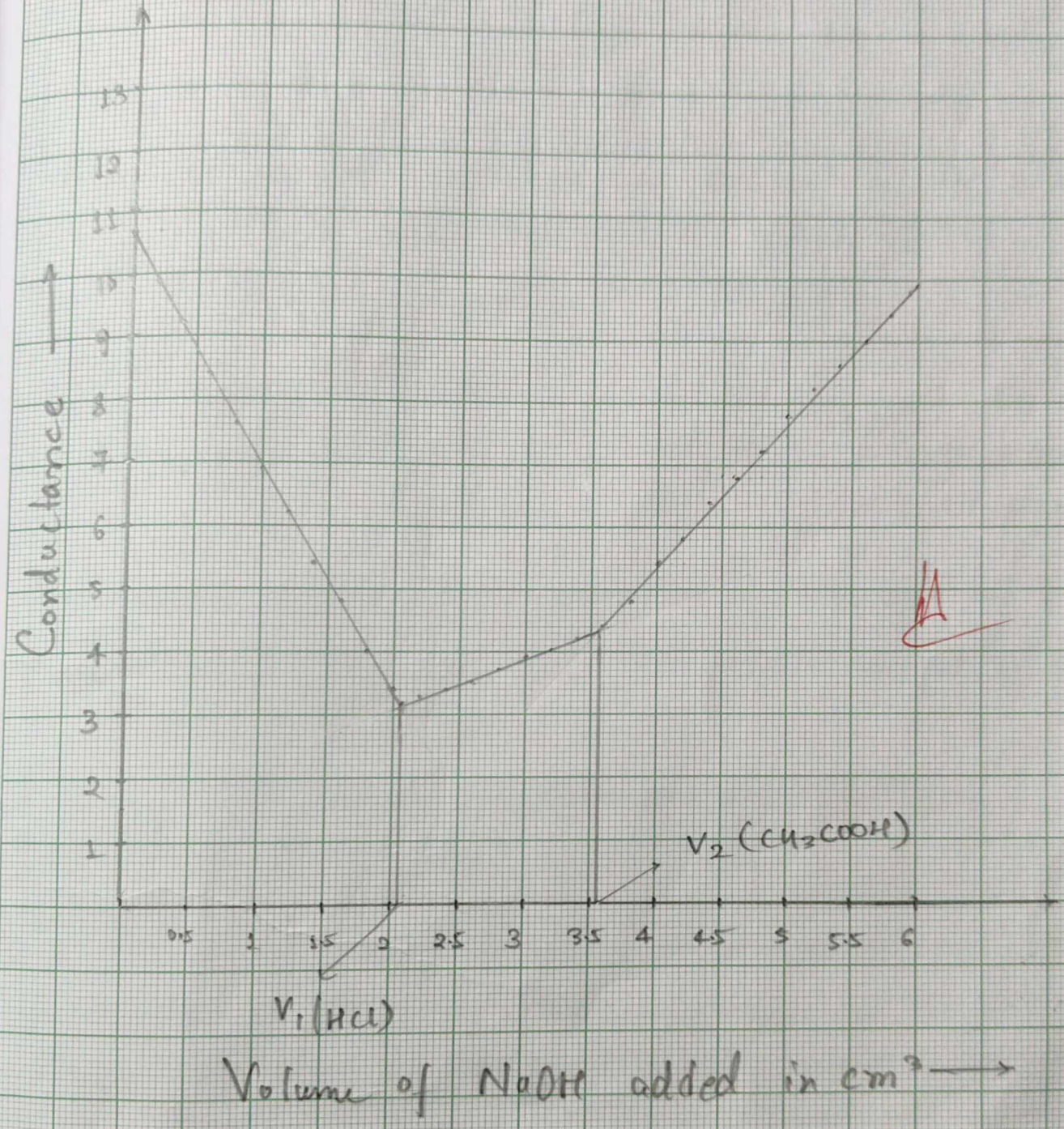
Volume of NaOH in cm^3	Conductance in mS	
0.0	10.68	10.7
0.2	9.91	9.9
0.4	9.15	9.2
0.6	8.43	8.4
0.8	7.63	7.6
1.0	6.89	6.9
1.2	6.16	6.2
1.4	5.39	5.4
1.6	4.83	4.8
1.8	4.04	4.0
2.0	3.43	3.4
2.2	3.28	3.3
2.4	3.42	3.4
2.6	3.53	3.5
2.8	3.71	3.7
3.0	3.88	3.9

Volume of NaOH in cm^3	Conductance in mS	
3.2	4.04	4.0
3.4	4.21	4.2
3.6	4.38	4.4
3.8	4.83	4.8
4.0	5.39	5.4
4.2	5.84	5.8
4.4	6.32	6.4
4.6	6.80	6.8
4.8	7.22	7.2
5.0	7.74	7.8
5.2	8.19	8.2
5.4	8.61	8.6
5.6	9.09	9
5.8	9.40	9.4
6.0	9.89	9.9

Scale:

$x \rightarrow$ 1 unit = 0.5 cm^3 NaOH

$y \rightarrow$ 1 unit = 1 unit Conductance



Calculation:

$$(NV)_{acid} = (NV)_{NaOH}$$

$$N_{acid} = \frac{(NV)_{NaOH}}{V_{acid}} = \frac{0.041 \times 2.05}{50} = 0.002N$$

Amount of HCl present in 1000 cm^3 of its solution
= N_{HCl} × gm equivalent weight of HCl
= $0.0020 \times 36.461 = 0.0729$ gm

$$(NV)_{CH_3COOH} = (NV)_{NaOH}$$

$$N_{CH_3COOH} = \frac{(NV)_{NaOH}}{V_{CH_3COOH}} = \frac{0.041 \times (3.55 = 2.05)}{50} = 0.00147N$$

Amount of CH_3COOH present in 1000 cm^3 of its solution = N_{CH_3COOH} × gm equivalent weight of acid.
= 0.00147×60.050
= 0.08827 gm

Inference: In this experiment, conductance of soln depends on no. of ions, change on the ions & also mobility of ions. HCl dissociates completely. After NaOH is added H^+ gets displaced by Na^+. Neutralisation of HCl, it will neutralise H^+ ions of CH_3COOH. There is a increase in conductance after same reading due to OH^- ions.

Relevance to Society & Environment:

Conductometric titration is used to determine water purity. It is used to check pollution level of water bodies like lakes, rivers, etc. It is also used to determine salinity of sea water & alkalinity of fresh water.

Report:

1. Normality of HCl = 0.00147 N, Normality of CH_3COOH = ~~0.00362~~ N
2. Amount of HCl present in 1000 cm^3 of its solution = 0.0729 g
3. Amount of CH_3COOH present in 1000 cm^3 of its solution = 0.08827 g

Evaluation of experiment - 6		
Components	Marks	
	Max	Obtained
Model Procedure, Model Graph & Calculation	9 + 7	16
Equivalence Point & Execution	16 + 4	19
Inference & Societal Relevance	2 + 2	3
Total	40	38
Signature of Teacher		17/11/23

EXPERIMENT NO – 7

Aim: Determination of viscosity coefficient of a given liquid using Ostwald's viscometer

Theory:

Viscosity arises due to friction between moving layers of a liquid. A liquid flowing through a cylindrical tube of uniform diameter is expected to move in the form of molecular layers. Layers close to the surface are almost stationary while that at the axis of the tube moves faster than any other intermediate layer. A slow-moving layer exerts a drag or frictional force on its nearest moving layer, backwards. This property of the liquid which retards or opposes the motion between the layers is called viscosity.

The coefficient of viscosity is defined as the tangential force per unit area required to maintain a unit velocity gradient between the two successive layers of the liquid situated at unit distance apart.

The coefficient of viscosity of a liquid is given by the Poiseuille's formula.

$$\eta = \frac{\pi P r^4 t}{8VL}$$

Where 'V' is the volume of the liquid, 'r' is the radius of the tube and 'P' is the pressure between the two ends of the tube and 'η' is the coefficient of viscosity. If equal volumes of the two different liquids are allowed to flow through the same tube under identical conditions, then

$$\frac{\eta_l}{\eta w} = \frac{t_l d_l}{t_w dw}$$

The time 't_l' taken by the given liquid to travel through a certain distance in the tube is determined. The time 't_w' taken by standard liquid (water) to travel through the same distance is measured. Knowing the densities of the two liquids (d_l and d_w) and also the coefficient of viscosity of the standard liquid, coefficient of viscosity of test liquid is calculated.

Description:

Take a clean and dry viscometer. Fix it vertically to a stand. Using graduated pipette, transfer known volume (say 10 cm^3) of the given liquid into the viscometer. Then it is placed in a water bath such that its big bulb is completely immersed in water. Suck the liquid above the upper mark of the small bulb of the viscometer. Allow it to flow through the capillary. When the liquid crosses the upper mark of the small bulb, start a stopwatch and when the liquid crosses the lower mark, stop the watch. Note down the time of flow in seconds.

Repeat the experiment 3-4 times and calculate the average time of flow.

Remove the liquid from the viscometer, clean well with acetone and dry it in oven. Cool the viscometer to room temperature. Similarly, measure the time of flow of water by taking the same volume. Calculate average time of flow of water. Note down the room temperature. Knowing the density, viscosity of water and density of the given liquid, viscosity of the liquid can be calculated using the formula given below.

$$\eta_l = \frac{t_l\, d_l\, \eta_w}{t_w\, d_w}$$

(1 milli poise = $10^{-4} Nsm^{-2}$)

Model Procedure /Flow Chart:

Take a clean and dry viscometer. The viscometer is fitted to a stand & is immersed in a waterbath such that its larger bulb is completely immersed in it. Fill the viscometer with the unknown liquid & suck the liquid slightly up to the upper mark of the small bulb of the viscometer. When the liquid crosses the upper mark, start the stopwatch. & when it crosses the lower mark stop it. Note the time taken by it. Similarly, repeat the process 3 times. After that the viscometer is washed with acetone & dried in an oven for 5 minutes. Then the similar process is done with water. for 3 time. Values for water is also taken & calculated for viscosity.

Model Calculation:

Room Temperature = T°C

Coefficient of viscosity of water = η_w

Coefficient of viscosity of liquid = η_l

Density of the water = d_w

Density of the liquid = d_l

$\therefore$ Viscosity of given liquid (η_l) = $\frac{\eta_w \times t_l \times d_l}{d_w \times t_w}$

where ($t_l \Rightarrow$ avg. time of liquid, $t_w \Rightarrow$ avg. time of water)

Observation:

Liquid	Time of Flow		
	in min: sec	in seconds	Average in seconds
Test liquid	1:25.73	85.73	t_l = 86.03
	1:26	86	
	1:26.37	86.37	
Water	1:56.13	116.13	t_w = 115.76
	1:55.03	115.03	
	1:56.13	116.13	

Calculation: lab temp. = 27.5°C ; density of given liquid = 0.8617
viscosity of water at 27.5°C = 8.5

$$\eta_{(l)} = \frac{t_{(l)}\, d_{(l)}\, \eta_{(w)}}{t_{(w)}\, d_{(w)}}$$

$$= \frac{86.03 \times 0.8617 \times 8.5}{115.76 \times 0.998} = 5.432 \text{ millipoise}$$

Inference: From the above experiment we can infer that viscosity of given liquid can be calculated by using Ostwald's viscometer and measuring time taken by liquid to flow from upper mark to lower mark. By using the given data viscosity of given data comes out to be 5.43 millipoise. Also, it is a temp. dependent characteristic.

Relevance to Society & Environment: Viscosity of a liquid is a factor used mainly in medical industry to determine the viscosity of cough syrup, etc. Gums should be highly viscous so that it has strong sticking action. It is also used in paint industry to make paints with proper viscosity.

Report: The coefficient of viscosity of the given liquid at 27.5°C = 5.43 $X10^{-4}$ Nsm^{-2}

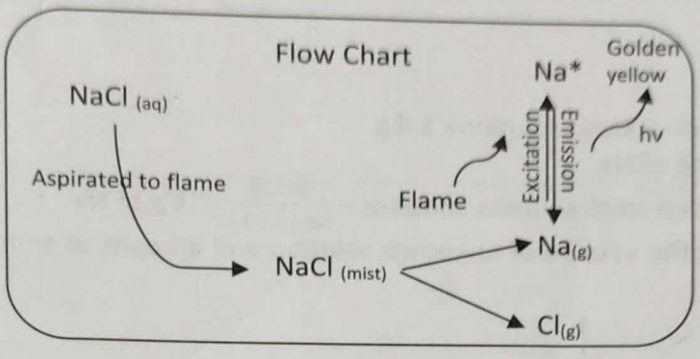
Evaluation of experiment - 7		
Components	Marks	
	Max	Obtained
Model Procedure, Model Graph & Calculation	7 + 5	12
Expected Value & Execution	20 + 4	24
Inference & Societal Relevance	2 + 2	3
Total	40	39
Signature of Teacher		10/11/23

EXPERIMENT NO – 8

Aim: Flame photometric estimation of sodium

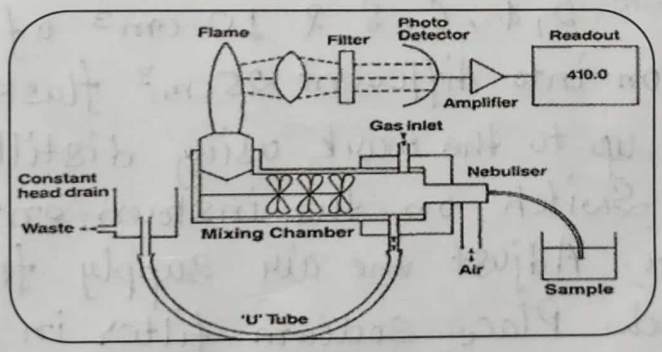
Theory:

Flame Photometry is an atomic emission technique used for detection of metals. If a solution containing metallic salts is aspirated into a flame, a vapour will be formed which contains metallic atoms. The electrons from the metallic atoms are then excited from ground state (E_1) to higher energy state (E_n) where n = 2, 3, 4, etc., by making use of thermal energy of flame. From higher energy states, these electrons will return to the ground state by emitting radiations ($E_n - E_1$) = hv where n = 2, 3, 4, etc., which are the characteristic of each element.



Flame photometer correlates the emitted radiations with the concentration of these elements. It is simple and rapid method for the elements that can be easily excited (Sodium and other alkali metals).

A flame photometer is composed of the pressure regulator, flow meter, an atomizer, burner, optical system, photosensitive detector and output recorder. A filter corresponding to the wavelength of light emitted by the element whose concentration is to be determined, is inserted between the flame and the detector. Propane gas is used as fuel and air or Oxygen is used as oxidant. Combination of these two will give a temperature of 1900 ^{0}C. The whole analysis depends on the flow rate of the fuel, oxidant, the rate of introduction of the sample and droplet size.



The sample containing the analyte is aspirated into the flame through nebulizer. Radiation from resulting flame is collected by the lens and allowed to pass through an optical filter, which permits only the radiation characteristic of the element under investigation into the photocell. The output from the photocell represents the concentration and nature of the element.

Description:

Transfer 2, 4, 6, 8 and 10 cm^3 of standard sodium solution into different 25 cm^3 volumetric flasks from the burette. Make up all the solutions using double distilled water. Stopper the flasks and shake well to get uniform concentration. To the given unknown solution, add double distilled water and shake well. Switch on the instrument; turn the gas supply on and light the gas at the burner. Adjust the air supply from the compressor to 10 lbs/sq inch using pressure regulator knob. Place the sodium filter (589nm) in position. Now dip the capillary tube in a cell containing double distilled water. The stream of air atomized as a fine mist draws up the liquid. Regulate the gas supply so that the colour of the flame completely turns to blue. Adjust the flame photometer to zero by means of zero control knob. Feed the various sodium solutions prepared, through the flame one by one including the unknown solution. Note down the flame photometer readings. Plot a graph of flame photometer readings against the volume of the solution get the calibration curve. Using the curve obtained find out the volume of the unknown solution containing sodium and calculate the amount of sodium in it.

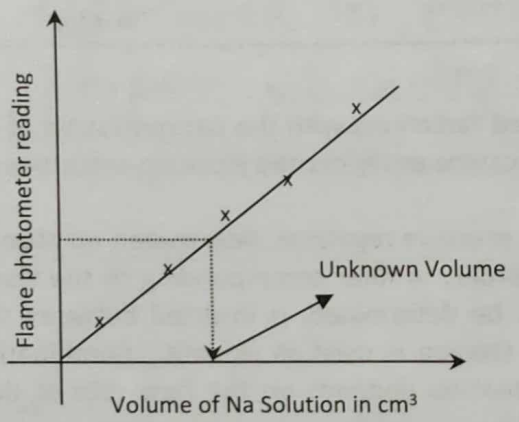
Calculation:

Amount of NaCl in the given 100cm^3 solution= 2.5 g

58.5 g of NaCl contains 23 g of Na.

Therefore, 1 cm^3 of the given stock solution contains = $\frac{wx23}{58.5x100}$ = 'A'g of Na

From the graph calculate the volume of unknown solution and amount of sodium present in the unknown solution.



Model Procedure /Flow Chart: 2, 4, 6, 8 & 10 cm^3 of standard Sodium solution into different 25cm^3 flasks from burette Make all sol^n up to the mark using distilled water. & Shake well. Switch on the instrument & light gas at the burners. Adjust the air supply from compressor to 10 lbs/sq inch. Place sodium filter in position. Now dip capillary tube in a cell containing distilled water Adjust the flame photometer to zero by means of zero control knob. Feed the various sodium sol^{n}s prepared, through the flame one by one including unknown sol^n. Note the reading Plot the graph & using it find vol^n. of unknown solution

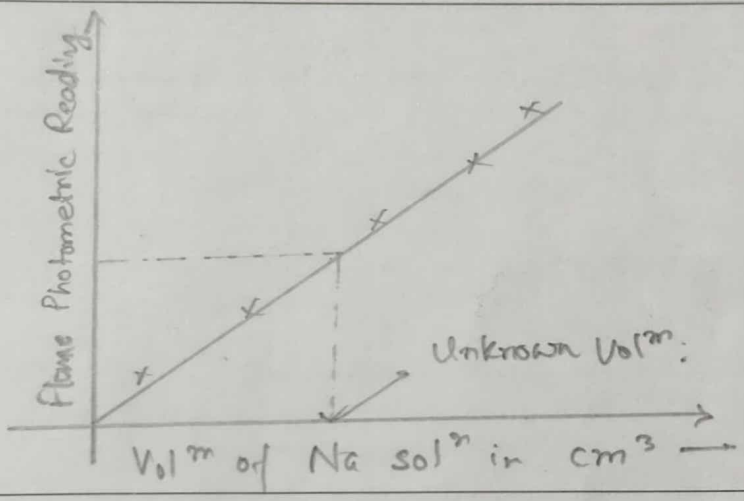
Model Calculation:

Amount of NaCl in given 1000 cm^3 solution = 2.5g

58.5 g of NaCl contains 23 g of Na

∴ 1 cm^3 of given Stock sol^n contains = $\frac{w \times 23}{58.5 \times 100}$ = 'A' g of Na ∴

Model graph:



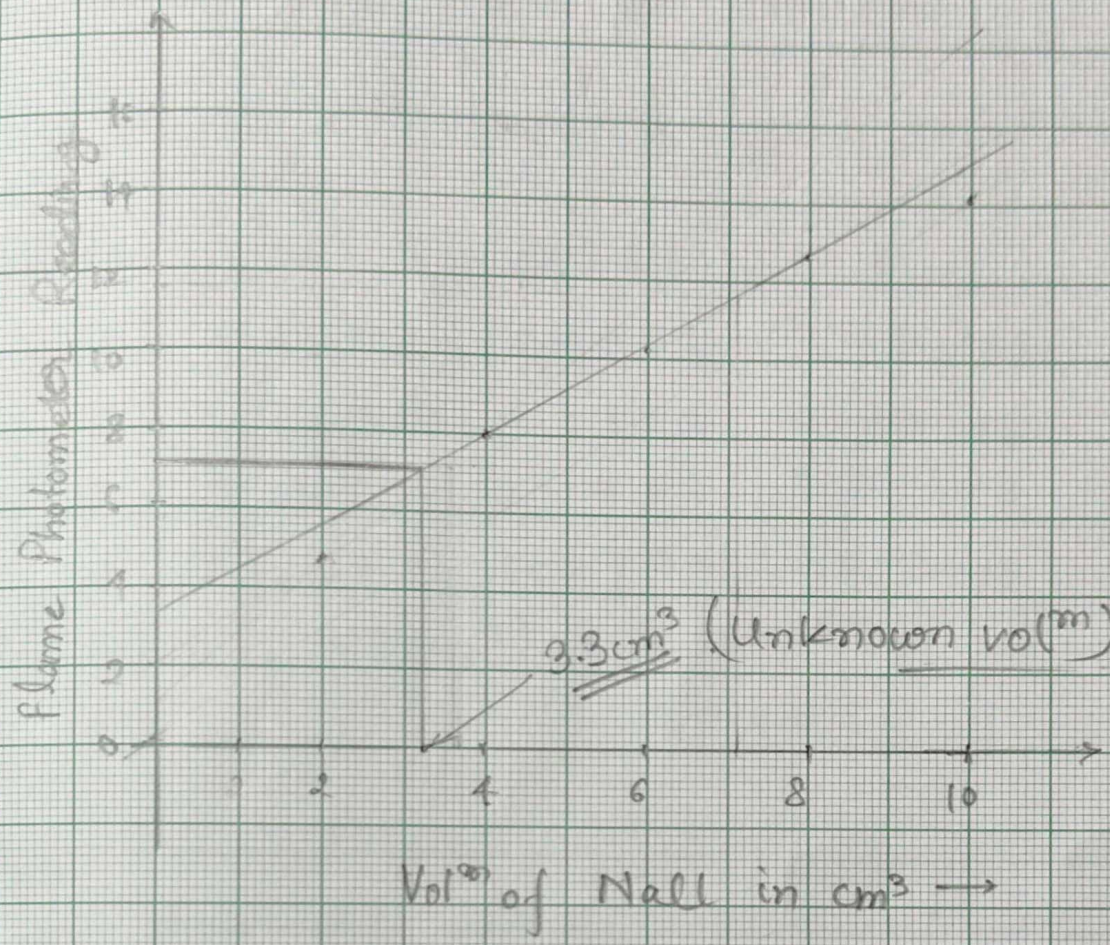
Tabulation:

Sl. No.	Vol. of NaCl in cm^3	Flame Photometer Reading	Wt. of Sodium in mg
1.	2	4.8	19.66
2.	4	8.0	39.32
3.	6	10.2	58.98
4.	8	12.6	78.64
5.	10	14.2	98.30
6.	Unknown	7.1	32.44

Scale:-

$x \rightarrow 1$ unit $= 1\ cm^3$ NaCl

$y \rightarrow 1$ unit $= 2$ unit reading



Calculation:

Amount of NaCl in the given 100 cm^3 sol^n = 2.5g

58.5 g of NaCl sol^n contains 23 g of Na.

$\therefore$ 1 cm^3 of given stock sol^n contains = $\frac{2.5 \times 23}{58.5 \times 100}$

= 0.00983 g of Na.

= 9.83 mg of Na.

$\therefore$ ① for 2 cm^3 ⇒ 2 × 9.83 = 19.66 mg

4 cm^3 ⇒ 4 × 9.83 = 39.32 mg

6 cm^3 ⇒ 6 × 9.83 = 58.98 mg

8 cm^3 ⇒ 8 × 9.83 = 78.64 mg

10 cm^3 ⇒ 10 × 9.83 = 98.30 mg

Unknown ($\therefore$ 3.3 cm^3) ⇒ 3.3 × 9.83

= 32.44 mg

Inference:

- Flame photometric ~~emission~~ estimation of sodium involves analysing the characteristic emission of sodium ions in flame.
- It is an atomic emission ~~technique~~ used for detection of metals.
- It excites the electrons from ground state to higher state which when return to ground, emits radiation, which gets detected.

Relevance to Society & Environment:

- This technique is commonly used for determining sodium level in diverse application.
- Some of these applications are- 1) Food analysis, 2) Environmental monitoring 3) Clinical Diagnostics.

Report:

1. Volume of unknown solution = 3.3 .... cm^3
2. Amount of Sodium in the given unknown solution = 32.44 ..... mg.

Evaluation of experiment - 8		
Components	Marks	
	Max	Obtained
Model Procedure, Model Graph & Calculation	9 + 7	16
Expected Volume & Execution	16 + 4	18
Inference & Societal Relevance	2 + 2	3
Total	40	37
Signature of Teacher		11/12/23

EXPERIMENT NO – 9

Aim: Colorimetric estimation of copper from PCBs

Theory:

When a monochromatic light of intensity I_0 is incident on a medium, a part of it may be absorbed (I_a), a part of it may be transmitted (I_t) and the remaining part may be reflected (I_r).

$$\text{Thus, } I_0 = I_a + I_t + I_r$$

I_r can conveniently ignore when the medium is transparent like perfect glass.

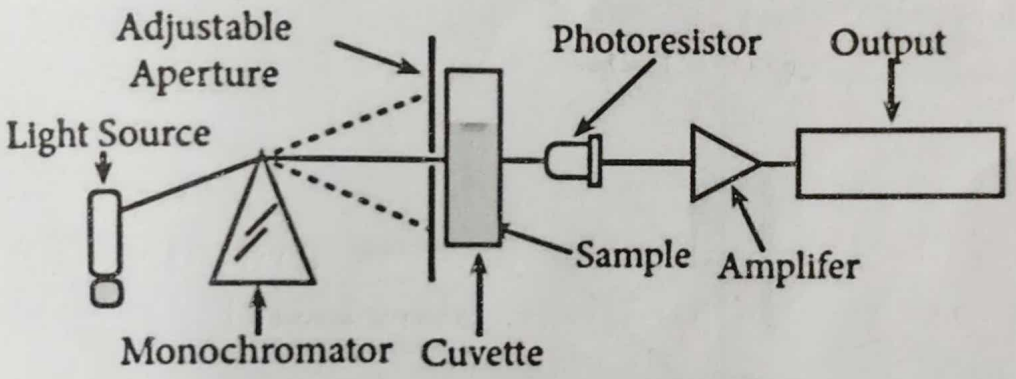
$$\text{Therefore, } I_0 = I_a + I_t$$

If I_t/I_0 is called transmittance then $\log (I_0/I_t)$is called absorbance or optical density (A). The relation between absorbance, concentration of the medium and path length or thickness of the medium is referred as Beer-Lambert's law.

$$A = \epsilon C l$$

Where 'A' is absorbance, 'c' is concentration, 'l' is thickness and 'Є' is a constant called molar extinction coefficient. Є is constant for a given substance.

Chemical analysis through measurement of absorption of light of a particular wavelength is known as colorimetry. The absorbance of light of a particular wavelength by a substance in solution varies directly with its concentration and the thickness of the solution. When the thickness of the medium is kept constant, the absorbance directly depends upon the concentration.



Instrumentation of Colorimeter:

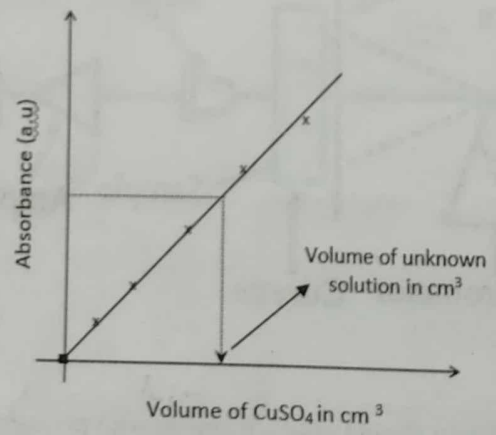
- **Light Source:** The source of light should produce energy with enough intensity to cover the entire visible spectrum (380-780 nm). Commonly, Tungsten lamps are used as a light source for measurement in the visible spectrum and near-infrared ranges. Halogen deuterium is suitable for measurement in the UV range (200-900 nm).
- **Slit:** It reduces unwanted or stray light by allowing a light beam to pass through.
- **Condensing lens:** Parallel beam of light emerges from condensing lens after the light passes through slit incidents on it.
- **Monochromator:** It filters the monochromatic light from polychromatic light, which absorbs unwanted light wavelengths and permits only monochromatic light. These are of three types: prism, grating, and glass.
    - **Prism:** It facilitates the refraction of light when it passes from one medium to another.
    - **Glass:** It selectively transmits light in certain ranges of wavelengths.
    - **Gratings:** These are made of graphite, which separates light in different wavelengths.

- **Cuvette (Sample cell):** The monochromatic light from the filter passes through the colored sample solution placed in the cuvette. Their sizes range from square, and rectangle to round and have a fixed diameter of 1cm. These are of three types based on the substances these are made of: Glass, Quartz, and Plastic cuvette.
    - **Glass cuvettes** are cheap and absorb light of 340 nm wavelength.
    - **Quartz cuvettes** facilitate entry of both lights of UV and visible ranges.
    - **Plastic cuvettes** are cheaper, easily scratched, and have shorter lifespans.
- **Photocell (Photodetector)**: These photosensitive devices measure light intensity by converting light energy into electrical energy.
- **Galvanometer:** The electrical signal generated in a photocell is detected and measured by a galvanometer. It displays optical density (OD) and percentage transmission.

Description:

Transfer 2, 4, 6, 8 and 10 cm^3 of given copper sulphate solution into different 25 cm^3 volumetric flask from a burette. Add 2.5 cm^3 of ammonia to all the flasks and makeup all the solutions up to the mark using distilled water. Stopper the flasks, shake well to get uniform concentration. Prepare a blank solution by taking 2.5 cm^3 of ammonia in another 25 cm^3 volumetric flask. Make up that solution also up to the mark using distilled water and shake well. To the given unknown solution (prepared from PCB sample) in 25 cm^3 volumetric flask also add 2.5 cm^3 of ammonia and make it up to the mark using distilled water and shake well. After 10 min. measure the absorbance of all solutions by using 620 nm filter of a colorimeter.

Note down the optical density of all solutions and tabulate the readings. Draw a calibration curve by plotting absorbance against volume of copper sulphate taken. Using the calibration curve, find out the unknown volume of copper solution given and calculate the amount of copper present in it.



Calculation:

Weight of copper sulphate pentahydrate present in 100 cm^3 of the given solution = 55 mg

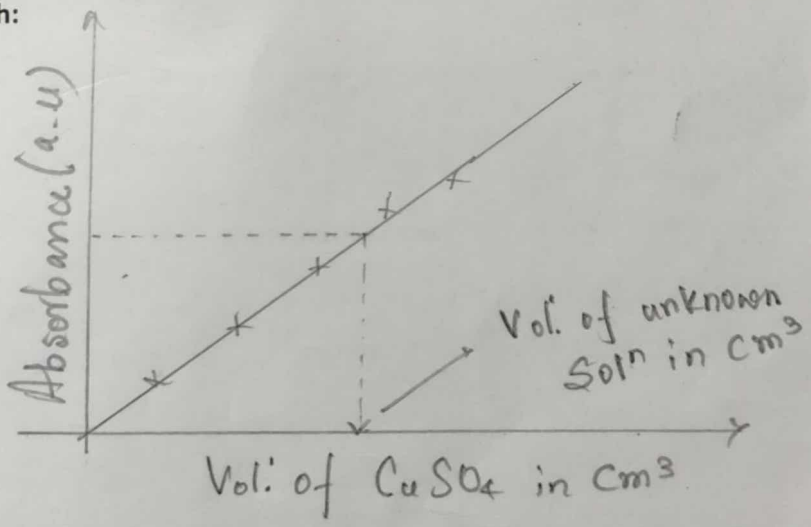
Weight of $CuSO_4.5H_2O$ in 1 cm^3 of its solution = X/100 mg =..... = 'Y' mg

Weight of copper present in 1 cm^3 of its solution = $\frac{'Y' * 63.54}{249.54}$ = 'W' mg

Model Procedure /Flow Chart:

Transfer 2, 4, 6, 8 and 10 cm^3 of the given copper sulphate solution into different 25 cm^3 volumetric flask. from burette. After adding 25cm^3 of Ammonia make all the solutions upto the mark by using distilled water. & shake well to get uniform concentration. Prepare a blank soln by taking 2.5 cm^3 of ammonia & filling rest of 25 cm^3 flask with water. To the given unknown solution in 25cm^3 flask add 2.5 cm^3 of ammonia & shake well. After 10 mins, observe all the solutions by using 620 nm filter of a colorimeter. Note all the readings & tabulate them. Draw the curve & get the unknown value of $CuSO_4$ by using the curve

Model graph:

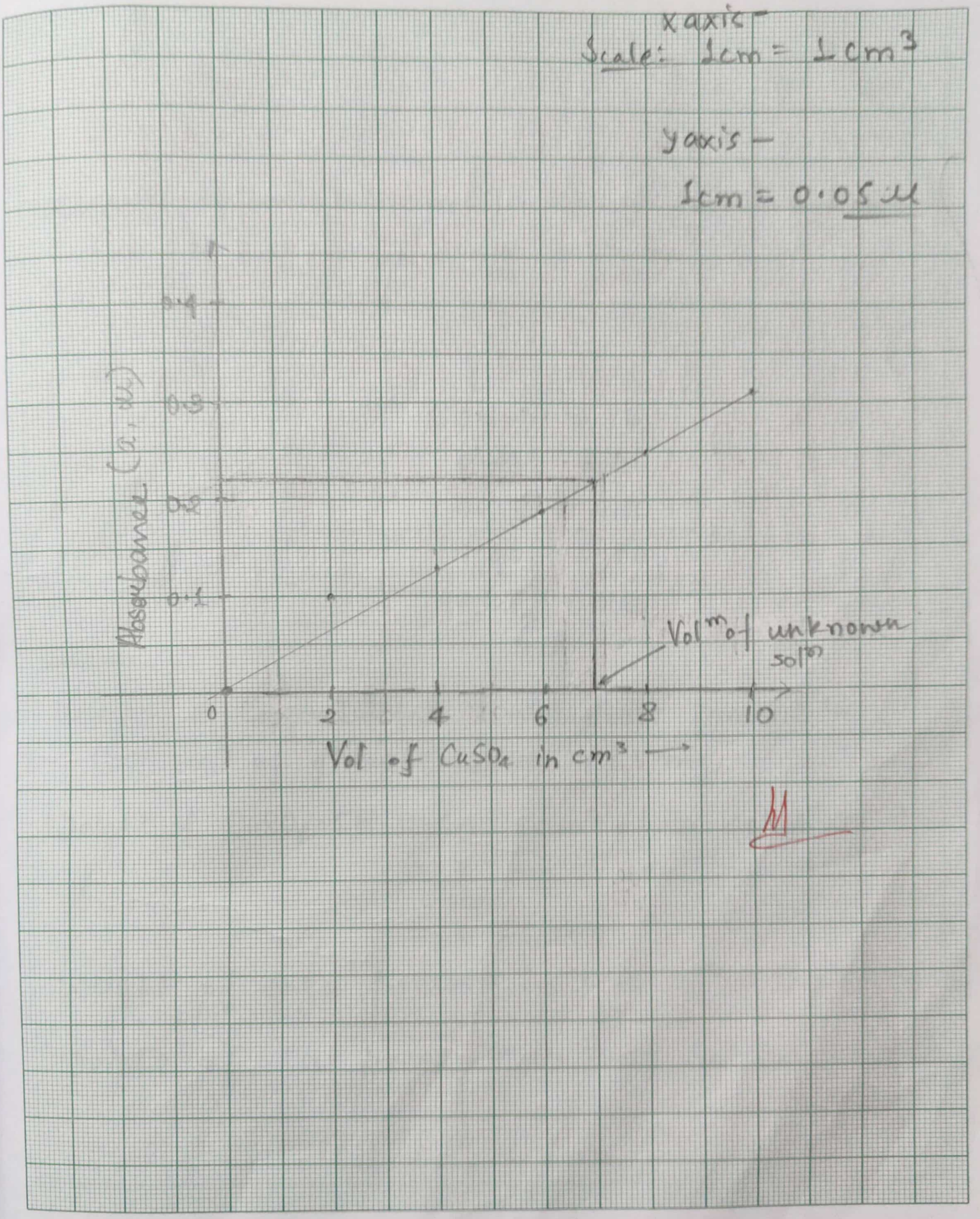


Model Calculation:

Weight of copper sulphate pentahydrate present in 1000 cm^3 of the given solution = 55 mg

Weight of $CuSO_4 \cdot 5H_2O$ in 1cm^3 of soln = X/100mg = Y mg.

Weight of Cu present in 1cm^3 of soln = $\frac{Y \times 63.54}{249.54}$ = 'W' mg


x axis –
Scale: 1cm = 1 cm³
y axis –
1cm = 0.05 u
Absorbance (in u)
0.4
0.3
0.2
0.1
0
2
4
6
8
10
Vol of CuSO4 in cm³ →
Vol of unknown sol^n

Tabulation:

Sl. No.	Vol. of $CuSO_4$ in cm^3	Optical Density	Wt. of Copper (mg)
1.	0 [Blank]	0.00	
2.	2	0.10	0.28
3.	4	0.13	0.56
4.	6	0.19	0.84
5.	8	0.25	1.12
6.	10	0.31	1.40
7.	Unknown	0.22	0.98

Calculation:

Weight of Copper sulphate pentahydrate present in 100 cm^3 of given solution = 55 mg

Weight of $CuSO_4 \cdot 5H_2O$ in 1 cm^3 of its solution

$$\Rightarrow \frac{x}{100} \text{ mg}$$

$$= 0.55 \text{ mg}$$

Weight of Cu present in 1 cm^3 sol^n = 0.140 mg

Weight of Cu present in test sol^n = 0.140 x

$$= 0.98$$

Inference:

The given $CuSO_4$ sol^n contains mg of Copper. Copper in its +2 state is blue in color. More the amount of Cu^{2+} present in the solution, more is the intensity of the colour.

Relevance to Society & Environment:

Using the principle of colorimeter we can create special containers that can be used for photosesitive materials. Colorimeter is the device used for testing the conc^n of sol^n by measuring its absorbance of specific wavelength. The filter on ~~Colorimeter~~ colorimeter might be set to red if liquid is blue.

Report:

1. Volume of unknown solution = ~~7.0~~ 7.0 .....cm^3
2. Amount of Copper in the given unknown solution = ~~0.98~~ 0.98 mg

Evaluation of experiment - 9		
Components	Marks	
	Max	Obtained
Model Procedure, Model Graph & Calculation	9 + 7	16
Expected Volume & Execution	16 + 4	20
Inference & Societal Relevance	2 + 2	3
Total	40	39
Signature of Teacher		1/12/23